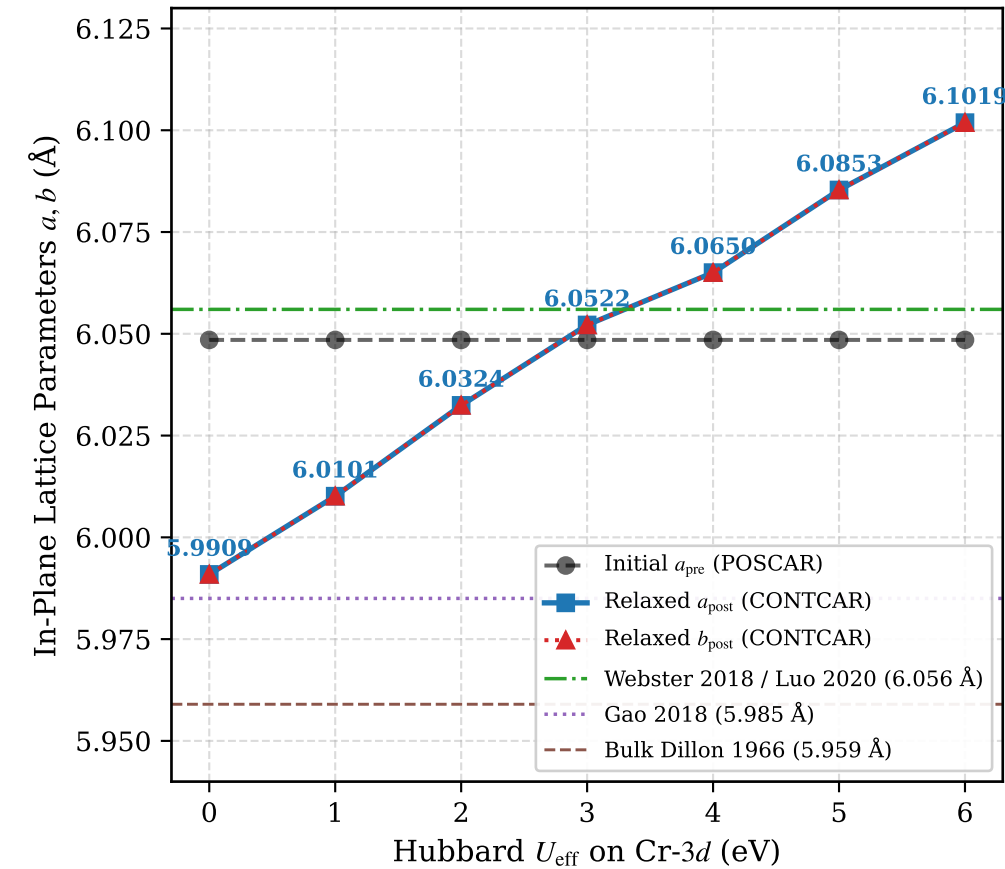
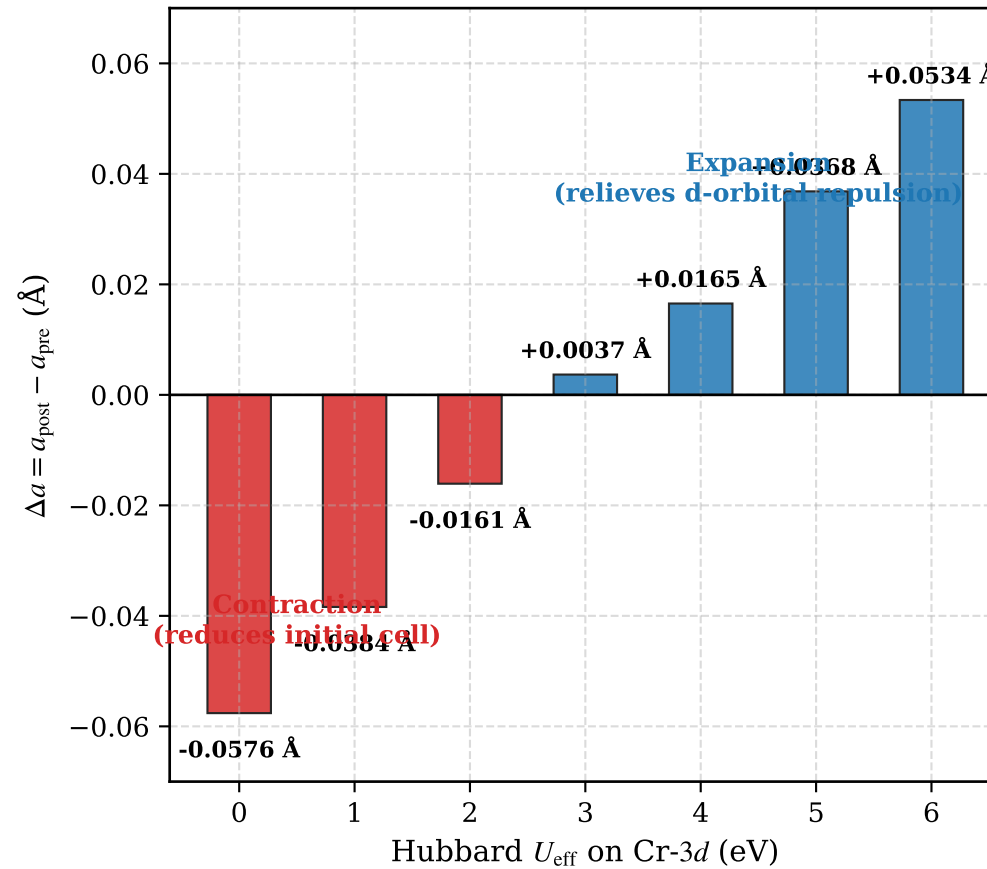


Monolayer CrCl_3 (1×1 Primitive Cell): Pre- vs. Post-Relaxation Structural Evolution

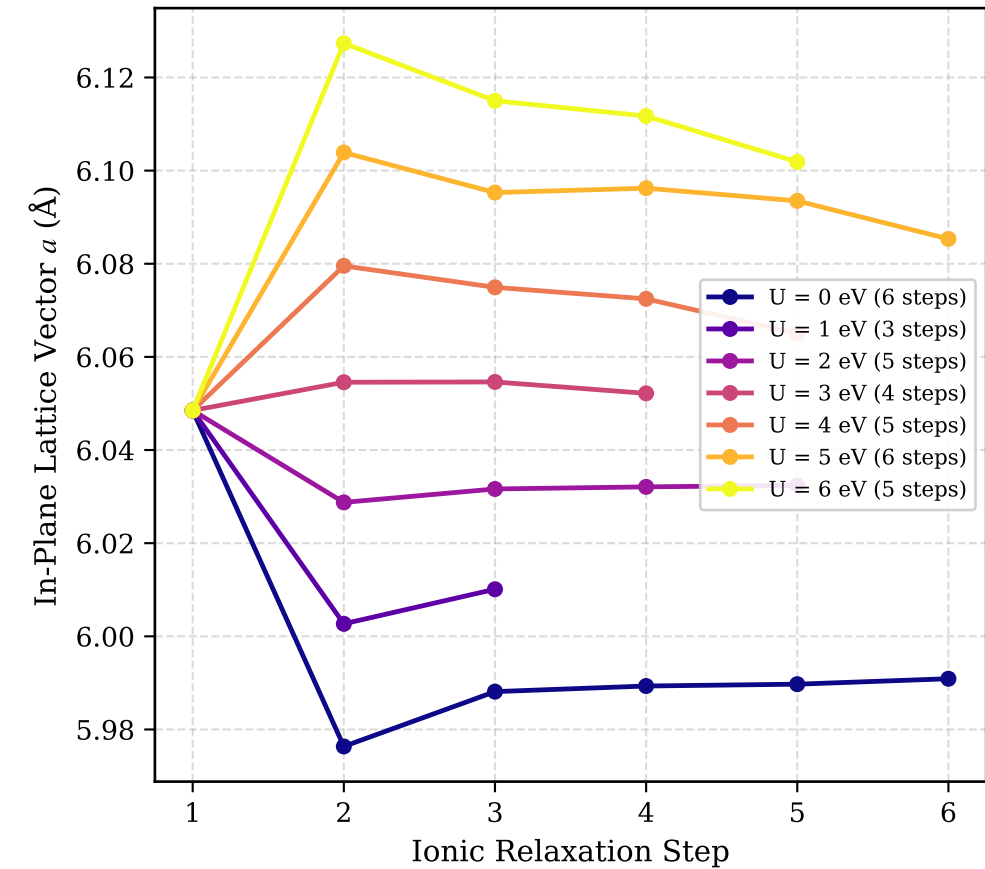
(a) Lattice a, b : Pre- vs. Post-Relaxation



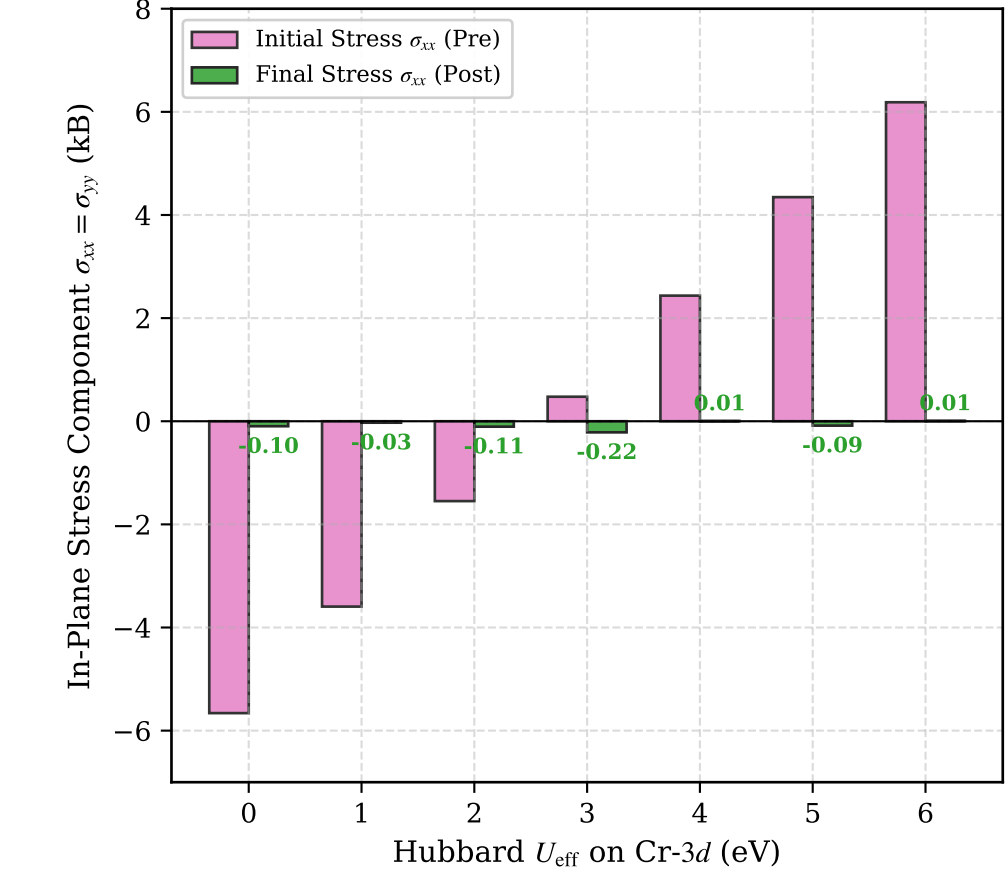
(b) In-Plane Vector Shift $\Delta a(U)$



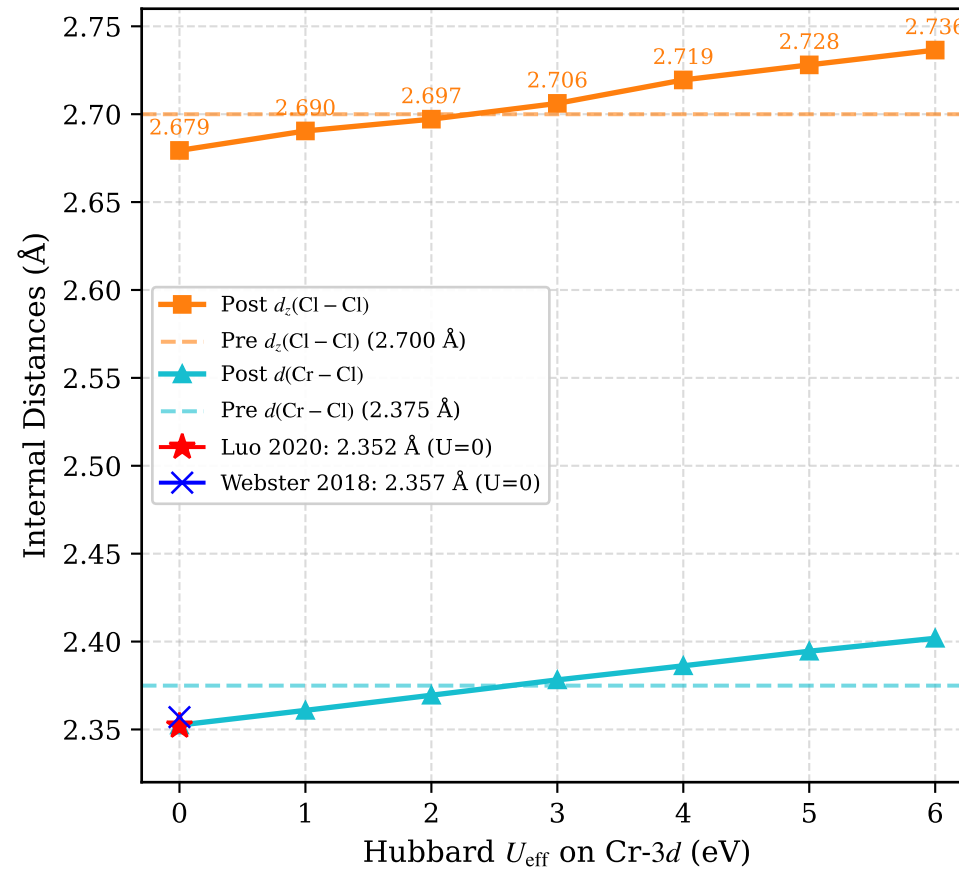
(c) Step-by-Step Trajectory $a(\text{step})$ from OUTCAR



(d) In-Plane Stress Minimization (σ_{xx})



(e) Vertical Thickness d_z & Cr – Cl Bond Length



Crystallographic Proof : Why $a \equiv b$ in Monolayer CrCl_3

- Space Group :** $P\bar{3}1m$ (No. 162), Point Group $D_{3d}(\bar{3}m)$
- C_3 Rotational Symmetry :** 3 – fold rotation axis $\parallel z$
 - $\rightarrow \mathbf{a}_1$ and $\mathbf{a}_2$ are symmetry – equivalent : $|\mathbf{a}_1| \equiv |\mathbf{a}_2| \Leftrightarrow a \equiv b$
 - $\rightarrow$ Inter – axis angle is strictly $\gamma = 120.000^\circ$
 - $\rightarrow$ Stress tensor is strictly isotropic : $\sigma_{xx} \equiv \sigma_{yy}$, $\sigma_{xy} \equiv 0$

Coordinate System Representation :

1. Hexagonal Primitive Cell (Our Work, Webster, Luo) :

$$\mathbf{a}_1 = (a, 0, 0), \quad \mathbf{a}_2 = (-a/2, a\sqrt{3}/2, 0)$$

$\rightarrow a = b \approx 6.056 \text{ Å}, \gamma = 120^\circ$ (2 Cr, 6 Cl atoms)

2. Rectangular Supercell (Alternative Representation) :

$$\mathbf{A} = \mathbf{a}_1 = (a, 0, 0) \rightarrow A = a \approx 6.056 \text{ Å}$$

$$\mathbf{B} = \mathbf{a}_1 + 2\mathbf{a}_2 = (0, a\sqrt{3}, 0) \rightarrow B = a\sqrt{3} \approx 10.489 \text{ Å}$$

$\rightarrow A \neq B$ (rectangular axes with $\gamma = 90^\circ$, 4 Cr, 12 Cl)

Conclusion : $a = b$ in our cell is NOT an error; it is a fundamental physical law of the honeycomb lattice!